

Systems Analysis and Design – INF2009F
Phumla Kamnandi Hotels Case Study – Workshop 4
Use Case Diagram and Use Case Scenarios (50 marks)

Please enter the names of students in your workshop group into this table

Student Name	Student Number	Project Group
1. Michael Copeman	CPMMIC001	30
2. Alex Cristaudo	CRSALE010	30
3. Shaylin Velen	VLNSHA004	30
4. Alex White	WHTALE015	30

Question 1 – Identify all the Actors and Base Use Cases in the System under Discussion (20 Minutes) [12 Marks]

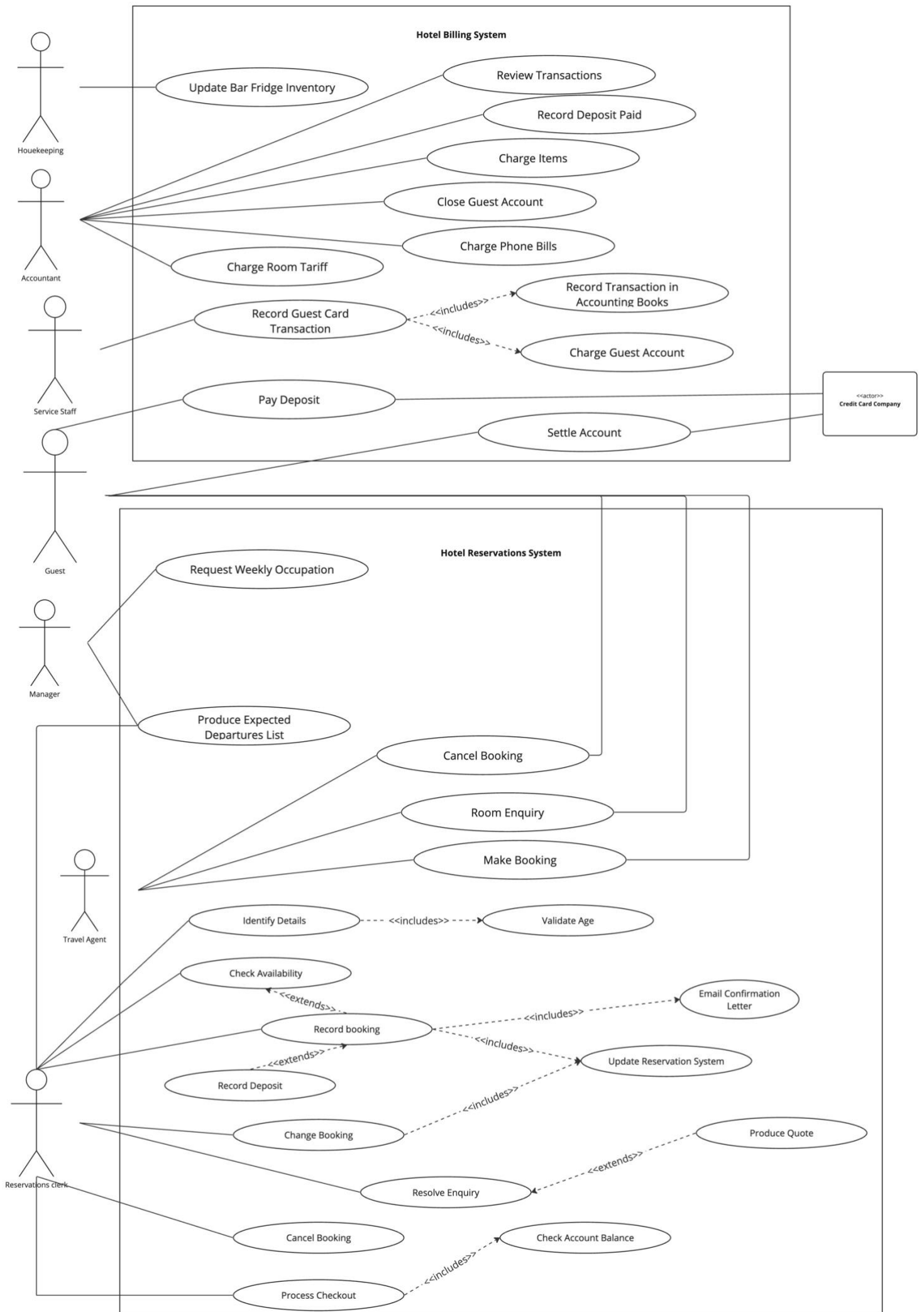
Populate the matrix below with all the actors and “base” use cases you would include in the Phumla Kamnandi Hotels Reservations and Billing System. Remember:

- **Actors:** are roles played by users of the system (remember this is always the user of the system and not the source of the data). If a customer can withdraw money from a bank (via an ATM) or approach a bank teller in the branch, both the customer and the bank teller would be actors in the “Make a Bank Withdrawal” use case. Remember we are not providing web based functionality in this phase of the project.

Use Cases: uses (*elementary business processes*) of the system. Ensure that each use case is a single task (could take more than one consecutive actions to complete) and not multiple tasks (clouds) or only part of a task (fish).

Actor	Use Case
Reservations clerk	Record a booking <<include email confirmation letter>> and <<include: update reservation system>> and <<extends: record deposit>> Check availability <<extends: record a booking>> Cancel booking Resolve enquiry <<extends: produce quote>> Identify details Change booking <<include: update reservation system>> Process check out <<include: check account balance>> Produce list of expected departures
Travel agent	Room enquiry Make a booking Cancel a booking
Housekeeping	Update Bar Fridge Inventory
Guests	Cancel booking

	Settle account Room enquiry Make a booking Pay deposit
Manager	Request weekly occupation Produce list of expected departures
Accountant	Review transactions. Record the deposit paid Charge items taken from bar fridge Charge phone bills Charge room tariff Close guest account
Service staff	Record transaction on guest card <<include: charge amount onto guest account>> and <<include: record transaction in accounting books>>



Question 3: Use Case Description (45 Minutes) [18 Marks]

The Use Case diagram you have just completed provides a very high level and easy to understand description of the functions performed by the Phumla Kamnandi Hotels Reservations and Billing system. However, in analysing these functions, you will have gained a more detailed understanding of the tasks that make up each business function. Rather than lose this important information about the system, you can document the detail using a use case narrative or scenario – basically a more detailed description of the inner workings of each use case.

Using the following template to assist you, develop a use case narrative for the Make a Booking use case.

In terms of the exceptional and alternative courses of events, brainstorm a list of at least 10 instances **and then** write up four of these in your use case narrative.

Expanded Use case:	Make a booking	ID: 0001	Level: High
Actors:	Guest Reservation Clerk Travel Agent Credit Card Company (secondary)		
Stakeholders and Interests	1. Accounting department - to record transactions related to making a booking 2. Advertising department - to evaluate the levels of occupancy 3. Management - to evaluate the performance of a hotel in terms of bookings		
Brief Description:	A guest/travel agent can make a booking either through email or phoning in. They give the dates they want to stay, along with the room package they want and the reservations clerk gives them a quote for their stay. The guest can then choose to reserve a room by making a deposit, or just make the booking by giving further information about themselves to the reservation clerk.		
Preconditions:	Guest must be 16 years old. Room must be available Credit card number provided by private individual Travel Agent confirm booking by email		
Post conditions:	The reservation system has successfully updated Deposit has been recorded (if a deposit paid) Confirmation email has been sent		
Related Use Cases	Includes: update reservation system, email confirmation letter Extends: Record deposit		

Typical Course of Events	
Actor Action	System Response
1.1. Guest/Travel agent phones to enquire about a booking	1.2. The system uses the reservation system to find available rooms and presents these options to the reception clerk who relays this message to the guest/travel agent
1.3 Capture guest details as well as details of their reservation	1.4 System updates to store guest information.
1.5. Guest/travel agent makes booking for the room they want	1.6. The system updates the reservation system to reflect this booking. The system will send an automated confirmation email
1.7.1. Guest gives their credit card number (if guests make booking) 1.7.2. Travel agent confirm booking by email and make deposit (if travel agents make booking)	1.8. The system will wait for approval from the credit card bureau, and then reflect this deposit on a guests account in the accounting system. The system will send out an automated email to confirm the deposit has been made
1.9. The guest arrives on the day of their booking and completes the necessary forms. The reception clerk then confirms and updates the system with this information	1.10. The system will reflect the guests room as occupied and store their expected date of departure

Exceptions / Alternative Course of Events	
Actor Action	System Response
2. 1 Credit card number given to the reservations clerk is invalid by the credit bureau. 2. 4 Ask the potential guest for a new credit card number. 2.5 Enter new credit card number	2.2 Remove payment method from system. 2.3 Follow up on the reason for card invalidity with the credit card bureau. 2.6 Enter new card number 2.7 Validate new card number with credit bureau else return to 2.4.
3.1 Deposit not paid within 7 days of booking	3.2 Change the booking to show no deposit paid. 3.3 Update reservations and room occupancy on system to allow the room to be taken.
4.1 There is no room available 4.2. The reservations clerk will ask the guest for a new date as their current date is invalid 4.4. The reservations clerk returns whether available or not. If not, return to 4.2.	4.3 The system checks for availability on that date and returns that to the reservations clerk

5.1 Customer wants to change booking details 5.2 Reservation clerk fetches customer details 5.4 Update necessary details on system	5.3 Fetch booking details from system 5.5 Save changes to booking and process on system/database 5.6 Update occupancy & reservation details or update user information
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List of 10 Exceptions or Alternatives to the Typical Course of Events	
There is load shedding at the hotel, and it has no power backup. Receptionist can take details by hand of potential guest and contact them when the system is back online.	
Credit card payment rejected	
Deposit not paid within 7 days of booking	
A guest asks to change rooms on the day of arrival - the reception clerk looks for availability and if it is available, updates the system to reflect the new room as taken and old as vacant	
Payment goes through but system does not record booking - ask for proof of payment and have booking reinstated into the system manually	
Guest calls but no one is at reception - is put on hold	
System is down under maintenance - Receptionist takes booking details and manually does booking when system is back online. If the system shows that the booking is not possible, the receptionist will call the guest to try and arrange alternative booking.	
There are no room available - the reservations clerk gives dates of available rooms	
Guest makes deposit but does not arrive on their day of arrival - the guest phones the guest back and looks for another date	
Deposit made in cash - the reception clerk treats this as a normal deposit without needing approval from the credit card company	